

Generative AI

Jio Institute PGP Program Artificial Intelligence and Data Science Year 2026-27

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Course Description

This course is the Generative AI foundation block of a two-part sequence (Generative AI, followed by a dedicated Agentic AI course). Generative AI represents a profound shift in the AI landscape - moving from task-specific models to modular, general-purpose systems capable of reasoning, retrieving, and generating context-aware responses. This course offers a practical, structured introduction to the GenAI stack: LLM fundamentals, prompt engineering, orchestration, and retrieval-augmented generation (RAG) - and closes with the essential building blocks of Agentic AI (tool use, simple agents) so students are not starting from zero.

Students in this program proceed to a 2-month internship immediately after this course. Session 7 has therefore been front-loaded with core agentic building blocks - tool/function calling and a simple tool-using agent - so students carry practical, internship-ready skills beyond RAG alone. The full depth of Agentic AI (multi-agent systems, memory, MCP, deep agents, coding agents) is covered in the follow-on Agentic AI course after the internship.

Learning Goals

- Build and debug applications using modular LLM components
- Apply prompt engineering techniques for reliable, controllable outputs
- Apply retrieval, embeddings, and indexing for context-aware responses
- Gain hands-on experience with LangChain and LlamaIndex
- Understand and implement basic tool-calling agents as a foundation for Agentic AI
- Collaborate on projects in a modern IDE (Cursor) with reproducible environments (uv)

Core Tech Stack

- **LLM orchestration:** LangChain
- **RAG:** LlamaIndex
- **Vector DBs:** FAISS, ChromaDB
- **Agentic workflows:** LangGraph (introductory)
- **Development environment:** Cursor IDE
- **Package management:** uv

Detailed Session Plan

Session 1 (7 September) | 3 hours | LLM Fundamentals & Prompt Engineering

- LLM Fundamentals: what makes AI “generative” vs. traditional ML; how language models work; what makes a model “large” (parameters + training data); a tour of frontier models (GPT, Claude, Gemini, DeepSeek, Llama)
- Tokens and the training process: self-supervised pre-training + fine-tuning, a brief look at RLHF, and a recap of the Transformer architecture
- LLM generation parameters: context length, rate limits; decoding/generation strategies (greedy, beam search, random sampling, temperature, top-k, top-p)
- Prompt Engineering: in-context learning (zero/one/few-shot), Chain-of-Thought and Least-to-Most prompting, prompt-injection risks (OWASP Top 10), and 10 practical tips for effective prompting
- Intro to LLM orchestration: the LangChain ecosystem
- **Hands-on:** First LLM calls; observe the effect of generation parameters; prompt-driven exercises

Session 2 (8 September) | 2 hours | LLM Orchestration + Embeddings

- LLM Orchestration: LangChain building blocks - prompt templates, chains (LCEL), output parsers
- Embeddings & Semantic Search: theory underlying retrieval - vector representations, similarity search
- **Hands-on:** Chaining prompts; first embedding-similarity exercise

Session 3 (9 September) | 3 hours | Foundations of RAG

- Recap: language models, LLM cost, Transformer architecture, generation parameters, and LangChain (RunnableSequence / LCEL)
- Introduction to RAG: the retrieval → augmentation → generation flow, and how it differs from chatting with an LLM directly
- Backbone of RAG - vector databases: chunking & embedding, similarity search, why embeddings work, and contextualized vs. static embeddings
- Are large-context models making RAG obsolete? Comparing RAG vs. large context models (LCMs) on cost, latency, long-context recall, flexibility, and security; the “needle in a haystack” test
- **Lab:** Retrieval Augmented Generation

Session 4 (15 September) | 3 hours | RAG Applications

- Recap: introduction to RAG and RAG vs. LCM trade-offs
- Comparison of popular vector databases: Pinecone, Weaviate / Milvus / Qdrant / pgVector, and FAISS
- **Lab:** Build and tune a practical, end-to-end RAG Q&A application
- **Assignment 2 context:** Build a RAG Q&A app with a Gradio UI where users upload PDFs and chat with them

Session 5 (16 September) | 2 hours | Advanced RAG

- Key failure points of a RAG system: missing content, poor ranking, context not reaching the LLM, extraction failures, wrong format, incorrect specificity, and incomplete answers
- Methods to improve RAG: query construction, query translation (decomposition, HyDE), routing, indexing strategies, retrieval ranking/refinement (Re-Rank, RankGPT, RAG-Fusion, CRAG), and generation-side active/self-RAG
- Agentic RAG: basic RAG chain vs. self-reflective RAG with LangGraph (retrieve → grade → generate, with query re-writing on irrelevant retrieval)
- **Lab:** Advanced Retrieval and LangGraph

Session 6 (17 September) | 2 hours | RAG Evaluations

- Why evaluate RAG applications: hallucinations, reduced reasoning, unexpected latencies, explainability, and building trust in high-stakes use cases
- Evaluating retrieval (“R”) vs. generation (“G”); who can be the “judge” - human, statistical NLP methods, or an LLM-as-judge
- Overview of evaluation metrics using RAGAs and DeepEval: Context Precision@k and Context Recall (retrieval metrics); Faithfulness, Answer Relevancy, Prompt Alignment, and Toxicity (generation metrics)
- **Lab:** RAG with LangGraph

Session 7 (21 September) | 3 hours | Finetuning LLMs & Multimodal RAG

- LLM observability and tracing (e.g., LangSmith) for debugging and monitoring LLM applications; how evaluation and observability work together
- Finetuning approaches: supervised finetuning, instruction finetuning (e.g., Alpaca), and their limitations; RLHF* (*not part of the exam)
- Parameter-Efficient Fine-Tuning (PEFT): how LoRA (Low-Rank Adaptation) drastically reduces the number of trainable parameters; hands-on with HuggingFace PEFT
- Knowledge distillation: deploying smaller specialized models (e.g., the DeepSeek-R1 distillation example)
- Prompt-based concepts: prompt compression and prompt caching
- Practical considerations: scale of training data, compute requirements, and the cost of training, using, and fine-tuning (full vs. PEFT) LLMs
- Multimodal RAG: unified vector space, primary-modality grounding, and separate-modality-store approaches, with their trade-offs
- **Assignment 3 context:** Multimodal RAG or Agentic AI exercise

Session 8 (22 September) | 2 hours | Building Blocks of Agents

- Introduction to AI agents: agent runtime (orchestration, memory - short/long-term, model-based reasoning/planning) and tools
- Tools and function calling: how LLMs interact with external tools/APIs, and key use cases (agents & multi-turn chatbots, batch inference, structured output generation)
- Levels of autonomy in LLM applications: code → LLM call → chain → router → state machine → fully autonomous
- The ReAct (Reasoning + Action) agent pattern, and other agentic AI design patterns: Tool Use, Basic Reflection, and Plan-then-Execute
- Multi-agent design patterns: Network, Supervisor, Supervisor-as-tools, Hierarchical, and Custom
- Comparing agentic frameworks: LangGraph vs. CrewAI vs. AutoGen
- **In-class exercise:** Designing agents for real-world domains (Retail, Tax/EY, EdTech, Healthcare Analytics, Government)
- **Lab:** Introduction to Agents with LangGraph

High-Level Summary:

Date	Hours	Focus	Outcome
7 Sep	3	LLM Fundamentals & Prompt Engineering	Understanding of LLMs, generation parameters & prompting techniques
8 Sep	2	LLM Orchestration + Embeddings	Chained LLM calls; similarity search
9 Sep	3	Foundations of RAG	RAG pipeline from scratch
15 Sep	3	RAG Applications	Practical RAG Q&A application
16 Sep	2	Advanced RAG	Improved retrieval techniques + Agentic RAG with LangGraph
17 Sep	2	RAG Evaluations	RAG evaluation using RAGAs & DeepEval
21 Sep	3	Finetuning LLMs & Multimodal RAG	Understanding of PEFT/LoRA, distillation & multimodal RAG

22 Sep	2	Building Blocks of Agents	Simple tool-using agent with LangGraph
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Evaluation

- **Assignment 1 - 20%:** LLM Fundamentals — Due 14 September 2026, 11:59 PM
 - **Assignment 2 - 20%:** RAG Applications — Due 20 September 2026, 11:59 PM
 - **Assignment 3 - 20%:** Group Mini-Project — Due 30 September 2026, 11:59 PM + Live Demos on 1 October 2026, 9AM
 - **30% Final Exam:** covers all concepts covered in class except the last session and RLHF concepts
 - **10% Class Participation:**
 - **(2%) Slido:** correct answers receive 1 mark and incorrect answers receive 0.5 marks
 - **(8%) 2 In-Class Coding Exercises:** hands-on exercises due the next day after class
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Class Participation and Attendance

You will find it extremely beneficial to attend all classes as a lot will be covered live and it'll be easy for you to keep up. You should inform the instructor beforehand about missed classes. The rules for missed classes are detailed in the student handbook provided by the Office of Academic Services.

Class Preparation and Conduct

It is presumed that you have access to a laptop computer for carrying out the work. Use of mobile phones in class is strictly not allowed. You should adhere to the code of conduct documented by the students' body. Plagiarism will be dealt with the strictest of measures.
